

Day 19 - Lecture 19.4: Box Plots (Five-Number Summary)

1. Overview & Purpose

A **Box Plot** (or Box-and-Whisker Plot), invented by mathematician **John Tukey** in 1977, is a standardized way of displaying the distribution of data based on a **five-number summary**:

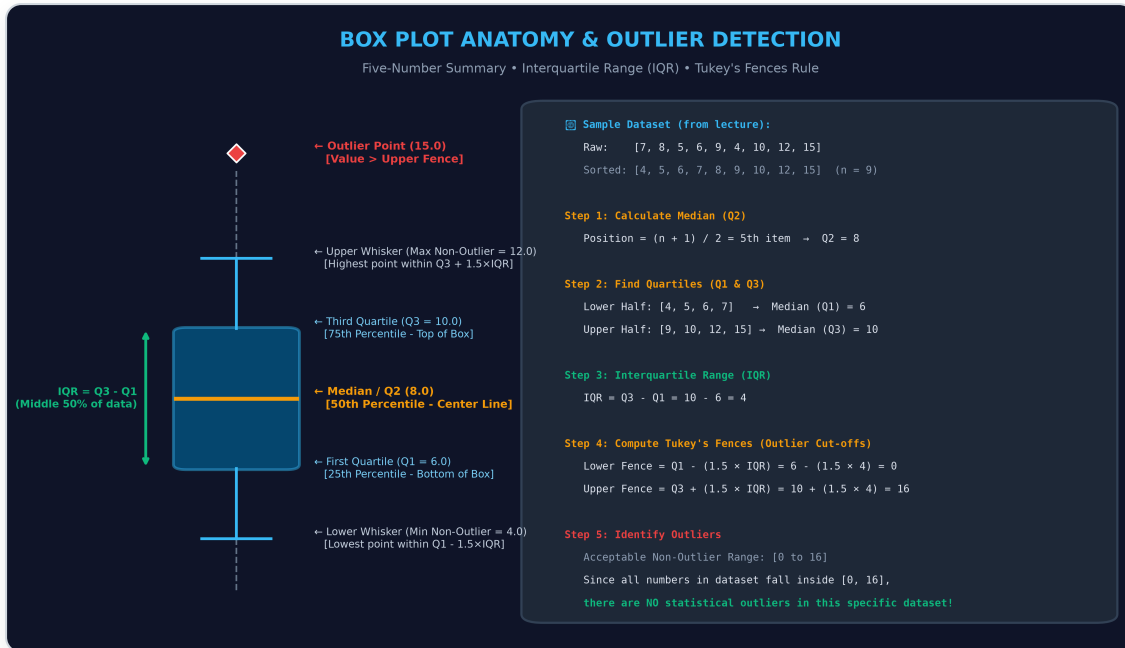
1. **Minimum** (lowest non-outlier)
2. **First Quartile (Q_1)** (25th percentile)
3. **Median (Q_2)** (50th percentile)
4. **Third Quartile (Q_3)** (75th percentile)
5. **Maximum** (highest non-outlier)

Additionally, it explicitly detects and plots **Outliers** beyond the whiskers.

2. Anatomy of a Box Plot



Visual Diagram & Derivation:



flowchart TD

```

subgraph BoxPlotAnatomy["Anatomy of a Box Plot"]
    A["Upper Outliers (> Upper Fence)"]
    B["Upper Whisker: Max within  $Q3 + 1.5 \times IQR$ "]
    C["Q3: 75th Percentile (Top edge of box)"]
    D["Median / Q2: 50th Percentile (Line inside box)"]
    E["Q1: 25th Percentile (Bottom edge of box)"]
    F["Lower Whisker: Min within  $Q1 - 1.5 \times IQR$ "]
    G["Lower Outliers (< Lower Fence)"]
end
A --- B --- C --- D --- E --- F --- G
  
```

Detailed Parts:

- Interquartile Range (IQR):

$$IQR = Q_3 - Q_1$$

Represents the middle 50% of the dataset.

- Lower Whisker Limit (Lower Fence):

$$\text{Lower Fence} = Q_1 - 1.5 \times IQR$$

- Upper Whisker Limit (Upper Fence):

$$\text{Upper Fence} = Q_3 + 1.5 \times IQR$$

- **Outliers:** Any point $x < \text{Lower Fence}$ or $x > \text{Upper Fence}$ is treated as an outlier and rendered as a distinct point (circle, diamond, cross).

3. When to Use a Box Plot?

- **Comparing Multiple Categories:** When you want to compare 5, 10, or 20 groups side by side. Histograms would get messy with 10 groups, whereas box plots sit neatly side by side.
- **Outlier Detection:** Immediate visual confirmation of extreme values.
- **Skewness Detection:**
 - If the median line is closer to Q_1 , the data is right-skewed.
 - If the median line is closer to Q_3 , the data is left-skewed.
 - If the median is centered and whiskers are equal, the distribution is symmetric.

4. Code Implementation

```
import matplotlib.pyplot as plt

# Sample dataset
data = [7, 8, 5, 6, 9, 4, 10, 12, 15]

plt.figure(figsize=(6, 6))
plt.boxplot(data)
plt.title("Basic Box Plot (Five-Number Summary)", fontsize=14, fontweight="bold")
plt.ylabel("Observed Values", fontsize=12)
plt.grid(True, linestyle="--", alpha=0.6)
plt.show()
```

5. Key Interview Questions

1. **Why $1.5 \times \text{IQR}$?** John Tukey chose 1.5 empirically. For a standard normal distribution, $\pm 1.5 \times \text{IQR}$ corresponds to approximately $\pm 2.7\sigma$ (covering 99.3% of the data). Points beyond it have a < 0.7 probability under normality, making them plausible anomalies.

2. **Is the median sensitive to outliers?** No, the median and IQR are **robust statistics** (resistant to extreme outliers), unlike the mean and standard deviation.